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# Artificial Intelligence Startups: An Empirical Study of Business Model Innovation in Digital Entrepreneurship

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***Abstract*—Prevailing digital innovation theory assumes fast go-to-market, rapid, and near-frictionless scaling of Software-as-a-Service ventures. However, artificial intelligence (AI) ventures developing data-driven solutions in the business-to-business (B2B) niche market tend to deviate from these assumptions. Drawing on a comparative multiple case study, we investigate how data dependency affects business model innovation (BMI) processes and trajectories in AI ventures. We contrast the early-stage business model experimentation of three AI digital ventures with that of three non-AI digital ventures operating in B2B niche markets. Our data corpus comprises 11 semi-structured interviews with 119 pages of transcripts and is complemented by informal interviews and archival data. Our analysis followed a transparent, process-oriented coding approach. The findings reveal three recurring limitations on business model experimentation from data-dependent solutions—data availability, epistemic predictability, and data path-dependency—which condition how BMI inputs are acquired, experimentation unfolds, and how scaling trajectories emerge. The AI ventures in our sample deviated from classical software as a service (SaaS) scaling patterns, instead relying on service-intensive launch strategies and staged automation to build a data corpus for AI training and expansion. By specifying data dependency as a structural constraint on business model experimentation and growth, this study refines core assumptions in digital innovation and BMI literature. It provides process-level explanations for slower and staged growth trajectories in AI ventures.**

***Managerial Relevance Statement*—Digital innovation emphasizes low development cost and rapid scaling after market launch for SaaS ventures. For entrepreneurs and engineering managers developing artificial intelligence (AI)-driven solutions, our findings indicate that these assumptions do not hold among AI ventures operating in data-scarce business-to-business niche markets. Our comparative study of three AI and three non-AI digital ventures shows that the data dependency of AI solutions influences how business models are experimented with and scaled over time. We find that AI ventures developing data-dependent solutions must navigate three key recurring tensions in early-stage business model innovation (BMI): acquiring sufficient and suitable data to commence model training, maintaining predictability and accountability when delegating decisions to AI, and navigating data path dependencies that constrain expansion across domains. For engineering managers and entrepreneurs, this implies that business model experimentation is interlinked with data acquisition and architectural choices in early-stage BMI. Early decisions and priorities regarding target market selection, pilot projects, and AI automation shape not only technical feasibility but also future growth prospects. Our cases illustrate how service-intensive launches can serve as bridges to generate revenue, accumulate training data, and validate business opportunities before committing to potentially irreversible growth paths—especially when operating in data-scarce business-to-business niche markets. These insights also help policymakers to adjust funding and startup support to staged AI development and prolonged experimentation rather than rapid scaling. Finally, this study addresses the United Nations Sustainable Development Goals 9 (Industry, Innovation, and Infrastructure) and 12 (Responsible Consumption and Production) by providing guidance for the accountable and resource-efficient development of AI solutions in data-scarce contexts.**

***Index Terms*—Artificial intelligence (AI), AI business models, digital artifacts, digital transformation.**

## I. Introduction

THE dominant view of digital entrepreneurship emphasizes short development cycles for digital products [1], automated product delivery and distribution [2], and rapid scaling of offerings across customers and industry segments [3]. Those findings reflect an established view in digital innovation research that attributes venture growth to modular architectures, the low marginal cost of replicating and modifying digital offerings, and distribution and monetization primarily through software-as-a-service business models [1].

However, initial studies on artificial intelligence (AI) business models document patterns of venture growth that deviate from this established view. For example, AI startups grow and scale more slowly than conventional software as a service (SaaS) startups and are structurally aligned to information technology service firms [1]. Previous research finds structural technical differences between AI and conventional software as a key inhibitor for AI adoption in firms [2], because the most advanced AI methods and algorithms depend on vast amounts of training data and training iterations. They are, therefore, costly to produce [3], [4], [5], especially in niche business-to-business (B2B) industries where data to train AI algorithms is scarce and fragmented, and requirements are unique to specific tasks and customers [6], [7], [8]. Moreover, AI ventures must navigate algorithmic opacity and evolving algorithm behavior in their product development efforts [9], [10]. Furthermore, the conceptualizations on business model experimentation and scaling might underestimate epistemic constraints stemming from uncertainty, requiring

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process- and context-sensitive explanations of how ventures experiment and learn over time [11]. Such insights indicate that there is an underspecified foundational issue in existing digital innovation theory: if and how the data dependency of AI offerings becomes a structural constraint to venture creation. Existing research defines embodiment and coupling of digital artifacts as the underlying structural constraints that affect new venture creation processes and their unfolding [12]. However, embodiment and coupling treat inputs to digital artifact creation in innovation processes as given. They do not explain how new ventures establish access to data as the precondition for artifact creation and evolution [7], [8], [13], [14]. Hence, we lack a clear understanding of how data dependency influences and constrains business model innovation (BMI) across process inputs, behavior, and outputs. This gap limits our knowledge of why AI ventures, unlike deterministic software trajectories, grow more slowly and adopt service-intensive business models. Hence, we address the following research question: How does the data dependency of AI digital artifacts impact inputs, behavior, and outputs of BMI processes in data-scarce B2B niche markets? Given the emerging nature of the underlying problem, we conducted a comparative multicase study of three AI digital ventures versus three non-AI digital ventures operating in data-scarce B2B manufacturing contexts. We conceptualized data dependency as a third structural tension alongside coupling and embodiment [12]. Data dependency manifests as three structural tensions—availability, predictability, and data-path dependency that shape BMI processes and explain why AI ventures pursue staged automation rather than rapid scaling.

## II. Digital Business Model Innovation and Digital Artifacts

Digital innovation theory assumes that digital ventures can achieve rapid market entry and growth through standardized products and infrastructure [3], [16], primarily in the form of software-as-a-service business models [1]. The explanations for such business model trajectories in digital contexts are the modularity, automation, and low marginal cost of replicating digital artifacts and solutions. However, these explanations are grounded in deterministic software artifacts that usually have low barriers to accessing the digital resources required to develop them. This raises questions about their applicability to the AI context—especially in data-scarce environments—where data is a critical input for developing data-driven solutions.

It is an established practice to study innovation and entrepreneurship as processes [15], [16], [17], [18], [19] to account for the dynamic nature of the underlying behavior, and the uncertainties innovators and entrepreneurs face at different stages of the process. A process view allows taking an agentic perspective of the individuals, the drivers of their actions, and how decisions and interactions with stakeholders shape early versions of their business model and product. For example, Andries et al. [15] found that new ventures experiment with up to six different business models before finding a successful one. Such experimentation is most promising for startups when conducted as rigorous hypothesis testing, as demonstrated by Camuffo et al. [16], thereby reducing the number of false-positive or false-negative decisions and leading to multiple changes in the business model along the way [17]. However, experimentation serves different functions along the new venture creation process, namely discovery-oriented experimentation to reduce epistemic uncertainty and making-oriented experimentation focused on generative artifact development [11]. As such, rapid iteration and scaling may not be uniformly applicable across digital innovation contexts. In the early stage of BMI, the early version of an intended product or offering, often referred to as a minimum viable product, is the focal artifact for experimentation and learning [18].

Digital BMI incorporates key concepts from the entrepreneurship process, such as opportunity recognition and exploitation under uncertainty, while also drawing on concepts from digital business, including data-driven value creation and control. Prior research highlights data network effects and data governance constraints to explain how the value and ownership of data shape market offerings and business networks [19], [20], [21]. These concepts provide explanations of value propositions, revenue models, and dominant design, rather than the process by which new business models form and emerge. From a process view, digital BMI is characterized by the speed at which digital artifacts evolve [22], [23] through continuous interaction of digital components, organizational arrangements, and external stakeholders [9]. Furthermore, digital tools are leveraged to accelerate business model experimentation [7], [13], [8], [24], [25]. Digital artifacts are inherently editable, reprogrammable, and distributable, making them generative and modular [26], which allows entrepreneurs to expose their MVP [27] to potential users, customers, or contributors and distribute the entrepreneurial agency throughout a community of inquiry [26], [28]. This combination of early exposure and co-creation turns early-stage digital artifacts into boundary objects between innovators and external stakeholders and challenges innovators to assess the readiness of their early product versions and the timing of their market launch. For example, Wessel et al. [29] examine prototype fidelity and find that medium-fidelity digital artifacts optimize resource acquisition and signal value to a specific target group, thereby linking the engineering practice of digital artifact development to business model innovation.

However, the composition of digital artifacts imposes constraints on innovation processes, as they can be embodied either as software (ephemeral), hardware (perpetual), or hybrid combinations. A second key characteristic of digital artifacts is their coupling. Building on Yoo et al.'s [30] concept of layered modular architecture, coupling refers to the degree to which components of digital artifacts depend on one another and require coordinated change to be executed, modified, or recombined. Loosely coupled artifacts allow independent evolution of components, whereas tightly coupled components require synchronized changes across layers and actors.

The composition of digital artifacts impacts BMI processes in their initiation, duration, and outcomes [12], and creates specific tensions in BMI (see Table I). For example, ephemeral components reduce resource requirements and allow rapid prototyping and quick market entry. Loosely coupled, ephemeral artifacts are iterated quickly, enabling fast adaptation after launch [27]. Much of the rapid global expansion of digital businesses can be attributed to the enabling role of cloud infrastructure to

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TABLE I
VENTURE CREATION PROCESS TENSIONS THAT ARISE FROM DIFFERENCES IN DIGITAL ARTIFACT COMPONENTS [12]

| Component property | Process dimension | | |
|---|---|---|---|
| | **Input** | **Behavior** | **Output** |
| **Embodiment** | **Tension of diversity** The degree of heterogeneity required of the inputs. | **Tension of intensity** The pace and concentration of the undertaken activities. | **Tension of pliability** The digital artifact's capacity for change and adaptation. |
| **Coupling** | **Tension of distributability** The distribution of component supply and functionality | **Tension of synchronicity** The temporal concurrence of activities needing coordination. | **Tension of generativity** A digital artifact's capacity for unprompted change via third parties. |

deploy offerings and validate market potential, enabling investors to fund explicit business model hypotheses [2]. Furthermore, modern digital tools enable entrepreneurs to form and test multiple hypotheses in parallel, leading to more product launches and increasing performance [31]. However, infrastructure and platform choices are deeply intertwined with BMI decisions, such as value propositions and revenue models, thereby influencing future pathways and constraining future choices [32]. For example, innovators and entrepreneurs aiming for a digital business model prioritize customization and modular offerings in the early stages to enable co-creation.

Such options are created by service-oriented infrastructures and decentralized controls (loose coupling), also referred to as architectural malleability. In contrast, tighter coupling and control are easier to maintain, optimized for performance, and can result in a defendable core logic that remains opaque to the outside world.

The emergence of AI adds another layer of complexity for early-stage innovators and entrepreneurs. The heavy reliance of AI artifacts on large amounts of high-quality data and iterative learning entails considerable costs but also creates new opportunities for value creation. Such learning algorithms have significant potential for business model innovation, enabling further automation and optimization of operations [34]. However, the development of AI artifacts faces significant challenges in data collection, data preprocessing, and computing infrastructure [5], [21], [33], [34]. Unlike deterministic software systems, AI systems learn from input data [35] to predict outcomes [38] and generate actions. This tight linkage between input data, predictions, and action raises questions about how established assumptions about modular architectures and generativity translate into AI-centric BMI processes. Platforms can play a decisive role in collecting the data needed for further algorithm training. For example, platforms that provide enough value to users allow innovators and entrepreneurs to collect even more context-specific data from user interactions [21]. However, this exacerbates the previously mentioned sufficiency constraint when large amounts or specialized data are needed to train algorithms to a level that provides enough value for users to use them.

Another challenge in AI development arises from algorithmic opacity [11], [13], which refers to the condition in which the algorithmic internal logic is hidden and the solution path to a specific prediction is incomprehensible or only partially comprehensible to nontechnical users and developers [8], [35]. As such, opacity can erode reputation, trust, and accountability, particularly if the predictions or actions of algorithms become incomprehensible or even discriminate against user groups [36]. Furthermore, such large-scale data collection and processing activities need to be embedded in a robust data infrastructure to optimize the learning process. Hence, a vital element of AI business model innovation is organizational transformation to enable AI-driven innovation [41], because traditional hierarchical structures often impede the agility needed for rapid AI adoption [2], [42], and additional complexities arise from algorithm reliability and integration into existing systems [2].

In summary, the existing model of von Briel et al. [12] provides a strong theoretical basis for studying the impact of AI artifact composition on BMI processes but does not explain why ephemeral AI startups scale less than conventional software startups—particularly in B2B contexts [1]—and why their value propositions range from simple data-driven services to complex outcome-oriented solutions [2]. This suggests a theoretical gap regarding the impact of AI-specific characteristics on business model trajectories.

## III. METHOD SELECTION AND DESIGN

We adopted an inductive approach using a comparative, multiple-case study design [43] to develop a theory of how data dependency shapes BMI processes in digital ventures. We chose a case study design to account for the emerging nature of BMI around AI offerings, where theoretical explanations for phenomena such as the observed differences between AI startups and conventional SaaS startups remain limited [1]. Furthermore, identifying and developing causal mechanisms requires rich processual data that captures sequential actions and decisions over time [37].

Our data analysis followed Gioia's [38] structured coding process, in which we iteratively transformed raw data (first-order codes) into second-order themes and, finally, into higher level aggregate dimensions (third-order concepts). This approach is suitable for inductive theory-building in an emerging, novel context.

Building on existing findings on how data dependencies and data scarcity limit innovation processes and trajectories, our objective was to determine whether these findings constitute empirical observations or impose structural constraints that shape the evolution of digital artifacts and, by extension, BMI processes [39]. To account for the dynamic nature of BMI processes and the evolution of digital artifacts, we guided our analyses by decomposing BMI into process inputs, behavior, and outputs [12].

### A. Case Sampling and Boundaries

One of our main assumptions is that existing theories on digital artifact evolution and digital BMI are only partially transferable to explain the empirically observed constraints of developing data-dependent AI solutions. To make these differences analytically visible, we followed a polar-type

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TABLE II
OVERVIEW OF DATA COLLECTION

| Data type | Data source | Quantity | Use in the analysis |
|---|---|---|---|
| **The first and second rounds of interviews and informal meetings** | • Navigator – chief executive officer (CEO) and chief financial officer (CFO) (both co-founders).<br>• Match – chief commercial officer (CCO) and co-founder.<br>• Classifier – CEO and chief technology officer (CTO) (both co-founders).<br>• Visualizer – CEO (co-founder) and senior developer.<br>• Provider – CEO and CTO (both co-founders).<br>• Validator – CEO and CTO (both co-founders). | • 11 semi-structured interviews with top venture executives totaling 119 pages of interview transcripts.<br>• Six informal meetings with the ventures (based on notes) to clarify timelines and sequence of actions and decisions (see Table VII in the Appendix). | • Gathering information to understand the ventures' incubation process, with a special focus on MVP development.<br>• Enrich the data via the second round of interviews; refine the semi-structured survey. |
| **Secondary data sources** | • Email exchange with informants.<br>• Company blogs and websites, pitches, news articles, and news interviews.<br>• Participation in startup events to learn more about the context and connect with investors and potential customers. | • Six emails.<br>• Four pitch videos.<br>• 15 news and blog articles.<br>• Three events. | • Triangulating data from the interviews with asynchronous responses to specific questions.<br>• Supporting and integrating interview data.<br>• Leveraging publicly available information to enhance the validity of the collected data. |

selection of cases [40]. Hence, we chose six extreme-contrast cases—three AI and three non-AI digital ventures.

All ventures operate in data-scarce, B2B manufacturing niches and developed ephemeral digital solutions, meaning pure digital solutions without specific integration to own or third-party hardware [12]. We defined AI startups as those whose core value proposition relied on the development of ML algorithms using customer-specific data, and non-AI startups as those that relied on rule-based, deterministic software.

To ensure comparability across cases, we applied the following case sampling criteria.

1) All ventures in the sample developed pure digital solutions without any embodiment in specific hardware.
2) All ventures are located in Southern Germany to mitigate regional or geographic differences (related to, e.g., funding availability).
3) To identify instances of extreme data scarcity, all sample cases operated in potentially data-scarce sectors. Five ventures operated in B2B environments and one in a business-to-consumer context, in niche markets.
4) As of this writing, all ventures are still operating to ensure a basic level of success (viz., survival).
5) All startups launched between 2018 and 2021.
6) All ventures received funding through public or private sources and engaged with pilot clients to ensure a basic level of legitimacy in the market.

## B. Data Collection

We collected data between March 2022 and November 2022, focusing on the ventures' initiation and actions toward developing a market offering. Data sources included semi-structured interviews, blog posts, social media posts, observations, and attendance at relevant startup events (see Table I). Secondary data sources were used to triangulate, enhance, and integrate the interview data, thereby increasing the validity of our interpretations. In total, we conducted 11 semi-structured interviews with founders and senior executives, resulting in 119 pages of interview transcripts. In addition, we conducted six informal interviews to clarify timelines and action sequences, which were documented in case-specific timelines (see Table VII in the Appendix). The interviews followed a semi-structured protocol to understand the venture background, product development, data collection practices, and BMI decisions (e.g., pivots). All informants provided consent, and all data were anonymized. Table II summarizes the data collection procedure.

## C. Data Analysis

We analyzed the data in MAXQDA (version 2022.8) to develop theoretical explanations for the differing BMI trajectories of AI startups and non-AI startups. Data analysis followed an inductive and iterative Gioia methodology [45].

In the first step, we mapped the chronological events for each venture using interview transcripts and supplementary materials. We generated detailed case memos to summarize key events, relationships, causal patterns, and insights potentially relevant to theory development. Table III provides a structured overview of each case, while Table VII in the Appendix summarizes the key events for each case and the provisional first-order codes. In the second step, we conducted a cross-case analysis using constant comparison [46] to identify similarities and differences across cases. Given the split between AI and non-AI startups, we paid special attention to negative evidence or themes that appeared only in one group. We generated a comparison matrix (see Table V) to summarize the initial data context of each venture, including practices for managing market and technical uncertainties, scalability intentions and limitations, and the implications of data scarcity for early-stage business models.

In the third step, we iteratively abstracted first-order codes into second-order themes and, finally, into aggregated dimensions. One author conducted coding. To ensure analytical rigor and consistency, emerging interpretations and coding decisions were regularly discussed among the authors during review meetings and as part of an overall validation procedure.

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TABLE III
CASE OVERVIEW

| Case code name | Provider | Navigator | Classifier | Visualizer | Validator | Match |
|---|---|---|---|---|---|---|
| **Technical core of initial offering** | AI | AI | AI | Deterministic software | Deterministic software | Deterministic software |
| **Project start** | 2019 | 2021 | 2021 | 2019 | 2020 | 2018 |
| **Market launch** | Nov 2023 | Jan 2023 | March 2022 | Apr 2022 | Mid-2020 | Feb 2020 |
| **Value proposition** | Enhance the robustness of ML models in manufacturing environments by generating synthetic data. | Allow users with motor impairments to navigate a web browser using alternative input methods (e.g., eye-tracking). | Reduce search time for required spare parts based on visual recognition. | Improve science communication. | Systematically search online shops for spare parts to provide price transparency. | Transparency into internal spare parts inventory. |
| **Funding stage (Nov 2022)** | Pre-Seed (Business Angel + public grant) | Pre-Seed (public grant) | Pre-Seed (public grant) | Pre-Seed (public grant) | Seed (VC) | Series A |
| **Industry** | Manufacturing and Automotive | Internet technology | Machinery construction | Research publishing and pharmaceuticals | Machinery construction | Machinery construction and automotive |

TABLE IV
CHAIN OF EVIDENCE FROM INITIAL CODES TO THIRD-ORDER CONCEPTS [41]

| 1st order codes | 2nd order themes | 3rd order dimensions |
|---|---|---|
| Data scarcity; Synthetic data generation; crowdsourced head/eye tracking data; user-validated image data; labelled data scarcity; capturing master data; medical abstract retrieval | Mechanisms to receive training data | Availability |
| Rule-based software; AI not feasible; progressive data collection; icon annotation bottleneck | Deterministic intermediary to accumulate training data | |
| High-confidence thresholds; Manual+AI validation; user validated classifications; manual edge case review | Human-in-the-loop data validation | Predictability |
| "Fake it till you make it" approach; iterative mock-up validation; closed group pilots; low-fidelity prototyping; wireframes; business model pivot to recommender; BM pivot from on-prem to SaaS; feature pull from customers | Business model validation through experiments | |
| domain-specific algorithms; Algorithm A/B testing; AI sandbox tests | Model hardening | |
| from service to platform; data donation through users; resource/capacity balancing; white-label b2b2c approach; Service projects funding product; consulting and service offerings; selected customer AI pilots | Provisional revenue models | Data-path-dependency |
| Stepwise automation roadmap; Gradual domain expansion; configurability trade-offs; internal low-code platform; AI open source versioning | staged architecture development | |

In the final step, we integrated the resulting themes into a BMI process model consisting of process inputs, behavior, and outputs [12] within the specific context of an AI solution [20]. This led to the identification of three aggregate dimensions—availability, predictability, and data-path-dependency—that capture structural constraints shaping early-stage AI BMI decisions and trajectories under data-scarcity conditions (see Table IV). Throughout our iterative validation procedure, we considered alternative explanations such as industry effects or firm maturity.

## IV. RESULTS

The results are structured along the three BMI process dimensions—inputs, behavior, and outputs [12]. Across these dimensions, we identified three process tensions related to data dependency—availability, predictability, and data-path-dependency—that structurally distinguish AI-ventures from non-AI ventures operating in B2B niche markets. The tensions resulted from our inductive analysis and are elaborated below.

### A. Process Input: Availability Tension

All six ventures in the sample started up with the intention of building data-driven services as digital offerings and scalable business models. However, all ventures faced acute data scarcity in the beginning. Some ventures had only unlabeled customer data, while others had no data and synthesized data for initial training. The three startups in the non-AI group initially considered developing an AI solution, but abandoned the idea due to unfavorable data prerequisites and pivoted to a deterministic software approach, as illustrated by the CCO of Match: "We didn't say on day one we are going to an AI solution … first trying to solve a processual customer problem without AI because of the lack of prerequisites." (CCO, Match). In contrast, the AI ventures invested in data pipeline development early on. Despite having only limited training data or synthesized data available, they proceeded with large-scale development before defining and training stable algorithms. "We are trying to bridge the data scarcity and data imbalance." (Founder, Provider). As

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TABLE V
SUMMARY OF RESULTS

| Case | Initial data context and acquisition strategy | Practices for handling technical / market uncertainty | Early monetization strategy | Scalability intent & limitations | Data-scarcity–business-model implications (AI versus non-AI) |
|---|---|---|---|---|---|
| **Provider (AI)** | No labeled production data initially; early training through synthetic images and time-series data. | Parallel testing of algorithms; pilots with selected customers; low-code shell for fast pilot execution | Project and consulting work; roadmap to SaaS | Gradual automation; client-specific training and adoption of AI models | Data scarcity shifted BM to a service model; continuous training of AI models to transition into SaaS |
| **Navigator (AI)** | Requires head and eye tracking data; initially crowdsourced data; limitation because of data privacy | Small-scale tests to collect ergonomic data; pre-training open source vision algorithm; On-device ML-modules | License fee | Vendor-agnostic software solution to run on diverse hardware | MVP based on a small proprietary data set; Freemium + user data collection for model improvement |
| **Classifier (AI)** | Small initial set of labeled spare parts; high data variance across different customer environments | Switched from automated selection to a recommender system with user validation; confidence-based release gate | Subscription per user account | Onboarding new customers requires data annotation | Data variance resulted in the rescoping of the value proposition |
| **Validator (non-AI)** | Web-scraped spare parts catalogs; unstructured customer ERP data | Chose deterministic matching algorithms due to insufficient training data for AI-first approach | Initial data cleanup and SaaS | Supply side scale with a growing number of spare part catalogues | Immediate SaaS due to the digitization of processes; later stage transition into AI development |
| **Match (non-AI)** | Manual labelling of data based on web-scraped pricing data | Human in the loop for validating critical results; build up a standard data scheme for later transition into ML | Platform fee + initial commission | Limited accuracy was gradually overcome with more data | Early data scarcity postponed AI; early value proposition focused on process digitization; transition into AI with more data |
| **Visualizer (Non-AI)** | Large amounts of unstructured abstracts of medical publications, manual icon tagging, and rule-based NLP | Preprocessing through subject matter expert (founding member); annotation through working students; incremental automation | Pay-per-visual-abstract-summary | Data annotation throughput | Annotation bottleneck made the venture go into a service-driven niche; immediate and sufficient customer value through deterministic offering |

part of building the right data management infrastructure, the AI ventures hired data engineers and annotators even before sales to accelerate data readiness. “Currently, unfortunately, we still rely on public datasets … that’s why we rely on datasets that already exist.” (CEO, Navigator). In contrast, the non-AI startups channeled their resources and developers into feature development and UX optimization. Furthermore, the AI founders developed interim revenue models, such as project fees and consultancy, to extend their runways. In contrast, non-AI ventures directly launched their offerings with subscription-based revenue models.

### *B. Process Behavior: Predictability Tension*

The AI ventures in our sample adopted iterative, low-fidelity experimentation to manage early-stage uncertainty in algorithm development. They tested multiple algorithms in parallel for the same task to compare performance across contexts and pilots. The non-AI startups launched beta versions and relied on incremental bug fixes rather than gated projects. To manage the risk of prediction errors, the AI ventures integrated a human-in-the-loop approach to double-check low-confidence results: "These marginal cases are currently also checked manually … we have set a high threshold, so a human has to go over that again. “ (CTO, Validator). This intermediate layer of checking was not required for non-AI startups. Following early algorithm experiments, the AI ventures initially selected simpler algorithms or rule-based approaches to generate an initial value for customers and only later adopted more data-heavy approaches like deep learning once they established their data pipelines. These ventures also forged data partnerships with customers to enlarge their available data pool in exchange for discounted pricing. “Our webcam eye-tracking prototype is still using multilinear regression, but Deep Learning has more potential when you have more data.” — (CEO, Navigator). Non-AI ventures built their solutions on stable technology stacks and offered discounted pilots to drive user adoption. Architecturally, AI ventures remained modular in the early stages and experimented with several algorithms for the same AI task and to support data generation. In contrast, the non-AI ventures focused on a relatively closed, cohesive architecture that was easier to maintain. “We have now achieved multiple MVPs … and built a low-code platform internally with the focus on data generation.” — (CTO, Provider).

### *C. Process Output: Data-Path-Dependency Tension*

Several AI startups in our sample realized that a highly automated offering could not be achieved in the short term, with initial datasets yielding lower accuracy and lower confidence in model outputs. Hence, they rescoped their value proposition from automation to decision support to address attainable accuracy and liability constraints. This pattern did not emerge in the non-AI startups, whose deterministic software features met desired reliability expectations without any human in the

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loop. "We don't sell it as a tool that takes the decision away; we sell it as a recommender, and the human is still responsible." — Founder, Classifier. As a result, the AI startups launched services and hybrid revenue models to validate their business hypotheses through early traction and cash flow while simultaneously expanding their data corpus and improving their AI algorithms. The perception of offering an AI solution opened alternative innovation-oriented budgets, often from outside the requesting customer department. "By just pronouncing the word AI, we get completely different options towards other budget pots." — Founder, Match. However, this had the disadvantage that such subsidy budgets were not intended for continuous funding of new solutions, but for testing new technologies. On the contrary, non-AI startups priced their solutions based on proven, immediate efficiency gains and workflow optimization.

The gradual optimization and introduction of increasingly complex AI models led AI startups to refine their core functionality and automate their offerings over time. Although this allowed startups to gradually move away from the human-in-the-loop approach, non-AI startups used this time to develop more advanced functions and features. This gradual approach slowed the AI ventures down, even though they launched MVP versions of their offerings, and achieving their initial value proposition took several development steps. To counter the perception of being too slow, these ventures emphasized the data flywheel effect, enabling them to build robust, flexible AI offerings that later led to higher automation and, in turn, higher profit margins. "Our models trained with data from other projects, and that creates scalability, transferring learnings from one project to the next." — (CTO, Provider). Summarizing our findings, the three tensions—availability, predictability, and data-path dependency—explained how data dependency in AI solutions conditioned BMI processes in early-stage AI ventures, operating in B2B niche markets. Our findings refine existing assumptions of digital innovation theory by specifying boundary conditions for entrepreneurial experimentation, new product development, and new venture growth in the context of AI. Table V summarizes these patterns and contrasts AI and non-AI with respect to the ventures' initial data context, uncertainty responses, monetization strategy, scalability intent, and BMI implications of data scarcity.

## V. Discussion

In this section, we discuss our findings and elaborate on the process-level tensions from data dependency in early-stage AI ventures. By examining these tensions across BMI inputs, behaviors, and outputs, we interpret how data dependency conditions early-stage BMI in AI ventures operating in B2B niche markets.

### A. Availability Tension

Existing theory of digital innovation assumes that, as soon as a founding team has the technical capabilities to develop software, they can begin developing actual software and enter the iterative experimentation process of new business creation [41]. von Briel et al. [12] capture this assumption in the input dimension as the tension of diversity, in which the increasing heterogeneity also increases the complexity of product development. The diversity tension implicitly assumes that the relevant inputs exist and are available.

The findings imply that this assumption does not always hold in data-scarce environments, as observed in data-scarce settings where AI development is delayed due to insufficient input data for model training. In these environments, data availability becomes a prerequisite for AI development and business model experimentation, rather than being merely a source of input heterogeneity.

Furthermore, availability constraints intersect with von Briel et al.'s [12] distributability tension, which concerns the challenge of allocating functionality across actors and modules, e.g., local or cloud-based functions. Required data for model training can be fragmented across actors (e.g., customer data and master data), with varying willingness to share data with others, e.g., due to perceived competitive risks [7], [42]. The activation of the required input requires data acquisition strategies to take legal and technical ownership of the training data. As a result, data acquisition strategies such as user-generated data, manual annotation, or data synthesis become critical design choices in the early-stage BMI processes of the AI ventures in our sample, shaping when and how experimentation could begin.

Hence, these findings indicate that data dependency has received limited attention in digital innovation theory, given its potential to condition the initiation of BMI processes.

### B. Predictability Tension

The tension of predictability stems from the probabilistic and adaptive nature of AI-based digital artifacts. Deterministic software programs produce rule-based outputs that can be verified through various test scenarios, e.g., unit tests. AI artifacts are based on probabilistic models, in which multiple subdecisions are made in succession until a final result is produced [35]. Hence, the overall decision logic of AI remains opaque. Furthermore, these artifacts are continuously trained on new incoming data, meaning they adapt their behavior and decisions to these new inputs [43]. The predictability tension requires accountability assessments postlaunch, which impose continuous coordination costs that exceed the typical pre-launch testing of deterministic software programs [34].

As such, the predictability tension goes beyond von Briel et al.'s [12] tension of synchronicity, which emphasizes coordination during development. Yet, the predictability tension also captures the postdevelopment accountability cycles in which entrepreneurs remain responsible for monitoring, validating, and overriding AI-driven decisions. These postdevelopment accountability cycles are necessary for AI startups because previous research shows that a lack of AI explainability leads to lower user adoption and trust and can hamper the integration of AI into business workflows [44]. Furthermore, this insight aligns with the previously mentioned work on discovery-oriented and making-oriented experimentation, because predictability constraints shift early experimentation towards discovery-oriented experiments such as hypothesis testing, human-in-the-loop validation, and staged automation, allowing ventures to transition into generative scaling. This synthesis helps explain why AI ventures in B2B niche markets rely on intermediary service-heavy bridges

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We add to the existing notion of coordination the aspect of "human-machine-interaction" as a behavioral tension toward achieving epistemic accountability over time [13]. As Shepherd and Majchrzak [20] suggest, entrepreneurs may delegate cognitive tasks to AI while retaining responsibility for judgments and outcomes. In AI ventures, this delegation is often mediated through human-assisted anomaly resolution, algorithmic confidence thresholds, and escalation protocols that shape both the design and commercialization of AI offerings, usually relying on service-oriented offerings in early business models [4], [34].

### C. Tension of Data-Path-Dependency

Existing literature on digital innovations emphasizes the scalability of digital offerings and business models [41], [45], [46], [50], enabled by standardized infrastructure [3], the replication of solutions across customer segments, and the low marginal cost of distribution. However, Lehmann and Recker [27] note that even startups with a deterministic software product invest heavily in software development after market launch to prepare the product and operations for scaling.

Our findings suggest that AI startups face an additional constraint: they require more training data to improve model performance and robustness, and to train models for additional uses. While both AI and non-AI ventures deploy complementary digital assets postlaunch, such as integrations or APIs to extend their offerings, AI ventures require additional domain-specific data to complement their offerings. In contrast, non-AI ventures rely primarily on code and configurations as complements. Substantially improving AI models requires vast amounts of new training data [4], thereby increasing the marginal cost of expanding into new domains.

Furthermore, ephemeral digital ventures engage in porting, adopting their core tech for new uses [27]. However, the AI ventures in data-scarce B2B markets typically need to adapt models and data pipelines to new customer environments. Hence, they enacted intermediary revenue models and traversed a cascade of consulting services and human-in-the-loop approaches, then stepwise automated their offerings before achieving scalable offerings.

Contrasting this finding with von Briel et al.'s [12] generativity tension—where modular code enables third-party extension—our findings suggest that data dependency constraints generativity in AI ventures. AI solutions may support API integration, but without the underlying data corpus and models trained on it, the integration is of limited value to the third party. This results in path-dependent scaling trajectories anchored on the evolution of the data corpus. To synthesize how these tensions unfold across BMI inputs, behaviors, and outputs, Table VI extends von Briel et al.'s [12] framework by incorporating data dependency.

The findings of our study provide ample practical insights for entrepreneurs, engineering managers, and policymakers who aim to integrate or enable AI solutions into BMI processes and operate in environments of data scarcity—particularly in B2B markets. In such conditions, data availability, predictability, and data path dependency impose structural constraints on business model growth and trajectories (see Fig. 1).

TABLE VI
EXTENSION OF TABLE I BY PROCESS INPUT, BEHAVIOR, OUTPUT TENSIONS DUE TO DATA DEPENDENCY

| Component property | Process dimension | | |
|---|---|---|---|
| | **Input** | **Behavior** | **Output** |
| **Embodiment** | **Tension of diversity** The degree of heterogeneity required of the inputs. | **Tension of intensity** The pace and concentration of the undertaken activities. | **Tension of pliability** The digital artifact's capacity for change and adaptation. |
| **Coupling** | **Tension of distributability** The distribution of component supply and functionality | **Tension of synchronicity** The temporal concurrence of activities needing coordination. | **Tension of generativity** A digital artifact's capacity for unprompted change via third parties. |
| **Data dependency** | **Tension of availability** The possibility of collecting sufficient and suitable input data for AI model training. | **Tension of predictability** The ability to predict the performance of an AI product in its real-world environment. | **Tension of data path dependency** The ability to launch an AI product that is suitable for distribution across diverse client environments. |

### D. Business Model Innovation and Experimentation

The identified tensions highlight the need to validate business models that rely on data-dependent solutions. This is necessary due to the additional costs associated with developing a data corpus to resolve availability tensions, improving the AI model postlaunch, and the higher costs of pivoting to a different idea due to data path dependency. Founders and engineering managers need to incorporate data-related hypotheses and milestones into their overall business model validation. As such, managers benefit from discovery-oriented experimentation before committing to irreversible and potentially premature automation and scaling decisions [11].

In practice, customer pilots and MVPs benefit from combining and testing hypotheses about data availability, model performance, as well as hypotheses about value propositions and revenue generation.

In data-scarce niche markets, founders and engineering managers may de-risk startup or new product failure due to wrong market timing [47] or value proposition mismatch with customer expectations [48], may launch a deterministic shell with the primary objectives to validate the business model and provide value for customers early on, before starting AI development. Collecting data through these early interactions allows ventures to transition to AI augmentation and eventually to AI automation of their offerings. As such, AI ventures launch services and consulting revenue models until their data corpus meets the requirements for SaaS revenue from a data-dependent solution. This type of stage-gate approach enables ventures to reduce false-positive decisions in their early-stage BMI process [16].

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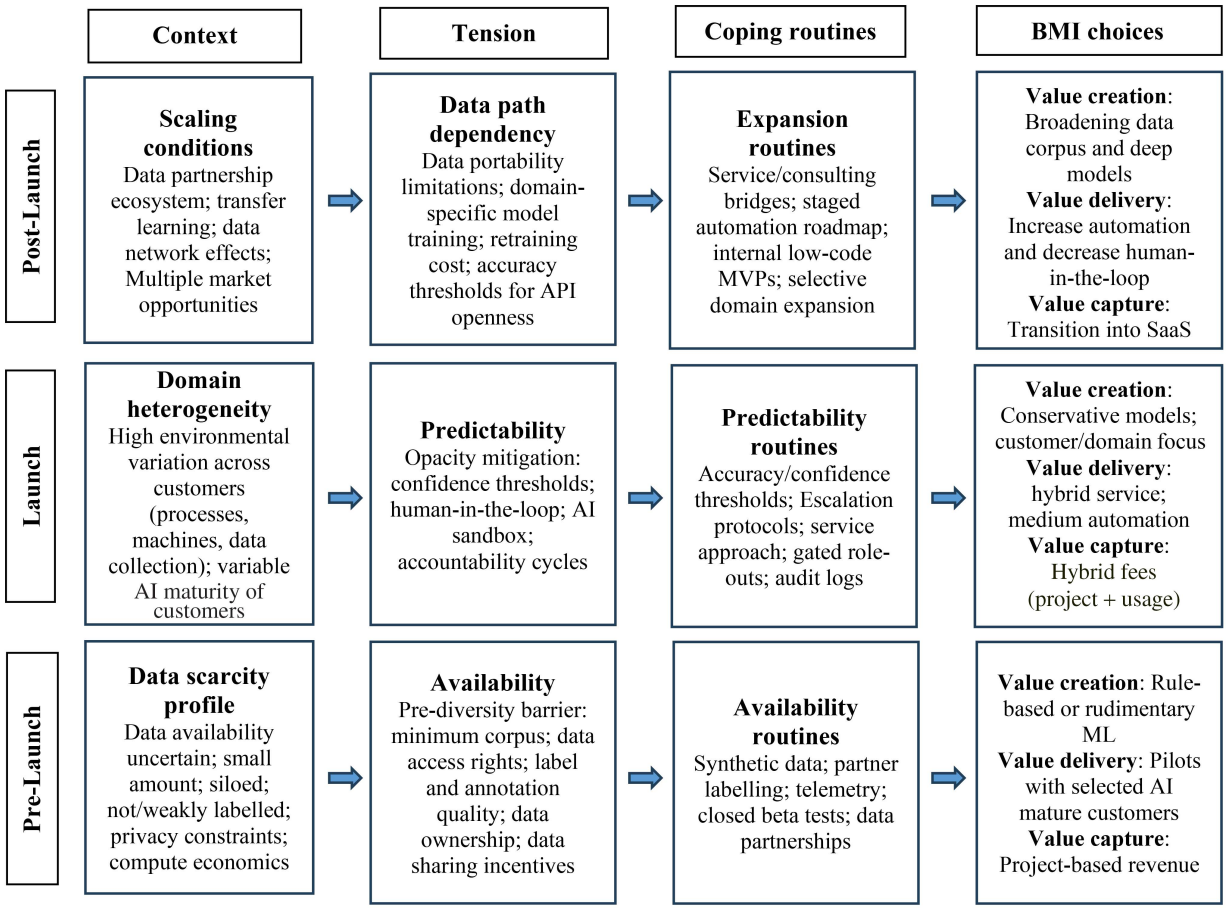



Fig. 1. Summary of venture maturity-based milestones and BMI choices from data dependency.

### *E. Managing AI-Specific Complexities*

Our research makes it evident that a data strategy is a vital element of engineering and product strategy for new ventures and incumbents aiming to develop AI solutions, particularly in data-scarce B2B niche markets. Therefore, founders and engineering managers should define data sources, data acquisition strategies, and data ownership early on to mitigate data availability bottlenecks [20].

When exposing their solution to customers, AI ventures benefit from augmenting human expertise with AI-driven decisions and from establishing confidence thresholds, escalation protocols, and override rules to manage predictability and accountability [34]. To maintain predictability over time, ventures should version datasets and models, monitor the accuracy drift of their AI models, and deploy reproducible data pipelines across use cases and customers [35]. Lastly, for privacy-sensitive AI applications, on-device AI learning can enable legal access to sensitive data and further reduce availability tensions [35], as illustrated in the navigator case.

### *F. AI Path Dependency and Structural Choices*

The data path dependency tension emphasizes the choice of the initial market opportunity and subsequent expansions into adjacent domains [49], [51], because AI development and model performance depend on accumulated, domain-specific data; such early choices shape growth and scalability in later stages. Hence, managers should sequence domain expansion to leverage transfer learning and to reduce the cost of retraining models for new uses.

Early in the BMI process, ventures should aim for reversible technical architectures, such as loosely coupled designs and modular data layers, to avoid early path dependency and reduce the cost of pivoting later [30]. At the institutional level, data partnerships and access rights are vital for meeting minimum availability thresholds and sustaining model development [49]. Lastly, to manage openness and generativity, APIs should be made available in a standardized form once AI solutions meet the reliability requirements within a domain, thereby aligning ecosystem extension and growth with the tensions of availability and predictability.

### *G. Implications for Policy Makers*

Our findings suggest that support needs for AI innovation in data-scarce B2B contexts go beyond funding algorithm development. Policies that, for example, include regulatory sandboxes or public–private partnerships may facilitate secure and legal data access, data-sharing infrastructure, and experimentation environments to overcome early availability and predictability constraints. Furthermore, funding measures that account for staged automation and service-based revenue bridges may better reflect our insights into AI-driven BMI than those that assume rapid scaling.

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Fig. 1 summarizes the implications of data dependency on BMI trajectories of AI ventures operating in data-scarce B2B niche markets over time.

## VI. Conclusion

We addressed recent calls to study digital artifacts [12] and how their development and evolution result in novel solutions and business models [8], [19]. Our inductive, comparative multicase study identified why established digital innovation theories do not fully explain the trajectories of business models in AI ventures. Across all cases, data availability, epistemic predictability, and data-path dependency repeatedly emerged as constraints to business model experimentation and scaling in AI ventures. These findings refine existing assumptions about digital innovation by specifying data dependency as a boundary condition for the experimentation and scaling of business models in data-scarce environments. They help explain why service-heavy product launches and staged automation are visible among early-stage AI startups. Given the inductive and process-oriented research approach, the findings are analytically generalizable but not statistically representative. The focus on Germany as the socio-economic context may influence data access conditions or funding structures. Furthermore, we focus on early-stage ventures and cannot capture how the three tensions evolve in later stages. Future research could test the identified tensions across larger samples, different domains, and develop quantitative measures to capture each tension.

## Appendix

TABLE VII
Case-Based Timelines and First-Order Concepts

| Case | Key events prior to market launch | Case-based 1st order codes |
|---|---|---|
| **Provider** | 2018-2020: PhD research on synthetic data applications in manufacturing automation; idea incubation<br>2021: Company founded; two founders; synthetic data MVP delivered in first pilot<br>2022: Multiple domain specific MVPs developed and delivered; low-code internal AI platform; switch from image-only to multi sensor data approach; high focus on service delivery vs product development (80/20)<br>Q4 2022: SaaS planned; resource shift from service delivery to product development (50/50) | Data-scarcity focus; synthetic data generation; from service to platform; domain-specific algorithms; selected pilot customer projects; resource/capacity balancing |
| **Navigator** | 2016-2020: PhD research on AI assisted web navigation with alternative means; idea incubation and grant application<br>2021: Start under public grant with four founders; reliance on public data sets<br>2022: Beta program; on device ML head and eye tracking; beta test to collect user-generated data and feedback loops | Data-scarcity (user generated data); privacy design (on device ML); beta testing to generate data and learn; configurability trade-offs |
| **Classifier** | 2021: idea to develop fully automated spare part identification; public grant application<br>2022: limited image data of spare parts; high data variance across customers due to edge cases -> value prop pivot to recommendation; ongoing data collection for ML training | Data scarcity; labelled data scarcity; value proposition pivot; user-in-loop recommendations |
| **Visualizer** | 2019-2021: 50+ customer interviews to validate problems in science communication; multiple mock-ups and "fake" chatbots to validate problem-solution-fit before coding program<br>2022: b2b2C launch with major science publishing house; semi-automated pipeline uses rule-based NLP; continuous transition into ML; hand-tagged icon database | Iterative mock-up validation; icon-annotation bottleneck; semi-automated data pipeline |
| **Match** | 2018: Idea emerged from founders' work experience; creation of an algorithm to match spare parts and inventory lists; AI not feasible due to data scarcity<br>2019: Pilot tests with a closed group of customers; pull-based development of new features (Global search, data lifecycle, parts-list checker) and release in pilot projects<br>2020: Start of data collection to build AI version of product | Rule-based product; AI not feasible; closed group pilots; progressive data collection; feature pull; Sequential roll-out |
| **Validator** | 2016-2018: Price intransparency for machinery spare parts identified through founders' work experience; web-scraping idea developed and incubated<br>2019: First pilot sold to machinery OEM before product was ready -> all weekend data cleaning (millions of rows) with 8 people; pivoted from on-premise to SaaS to become data provider; more datapoints added (availability and delivery time<br>2020-2022: Google-blocking incident required to build robust web-crawling infrastructure; iterative ML development (decision tree + manual validation) | Fake it till you make it-approach to validate business opportunity; feature pull form customers; Manual + AI hybrid; BM pivot from on-prem to SaaS; web-data uncertainty; human in the loop |

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